Combinatorics of the Poincaré Transport Residual: Discretisation, Monodromy, and Cylindric Partitions

Speculative notes from a conversation

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Abstract

We sketch a network of connections between the Poincaré transport residual ρ_T of a viscous elliptic structure (VES) and the combinatorics of exactly solvable lattice models. The central observation is that the rigidity condition $G = 0$ is a continuous analogue of the Yang–Baxter equation, and that ρ_T measures distance from exact solvability. When discretised on a lattice, the VES compatibility condition yields local rules of the type appearing in growth diagrams, puzzle pieces, and transfer-matrix formulations of Schur functions. We propose that the profile of a cylindric partition is a discrete monodromy, the combinatorial counterpart of ρ_T.

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1 The Poincaré transport residual

Let $\Omega \subset \mathbb{R}^2$ be a domain. A *viscous elliptic structure* (VES) on Ω is specified by a smooth generator $i\colon \Omega \to \mathbb{D}$, where $\mathbb{D}$ is the open unit disk. The associated Cauchy–Riemann operator

is

$$\partial_{\bar{z}_i} f \;=\; \tfrac{1}{2}\big(f_x + i(x,y)\, f_y\big),$$

and a function f is *i-holomorphic* when $\partial_{\bar{z}_i} f = 0$.

Definition 1 (Rigidity obstruction)**.** The *obstruction tensor* of a VES is

$$G \;=\; i_x + i \cdot i_y.$$

The structure is *rigid* (admits a full holomorphic calculus) if and only if $G = 0$.

Definition 2 (Poincaré transport residual)**.** The *Poincaré transport residual* is

$$\boxed{\rho_T \;=\; \frac{|G|}{1-|i|^2} \;=\; \frac{|i_x + i \cdot i_y|}{1-|i|^2},}$$

where $(1-|i|^2)^{-1}$ is the conformal factor of the Poincaré metric at $i \in \mathbb{D}$.

Key properties:

(i) **Invariance.** ρ_T is invariant under Möbius automorphisms of $\mathbb{D}$ acting on i—it is a genuine geometric invariant.

(ii) **Blow-up.** As $|i| \to 1$ (degenerate ellipticity), $\rho_T \to \infty$ even for bounded G, correctly detecting the parabolic/hyperbolic boundary.

(iii) **Vanishing.** $\rho_T = 0$ if and only if $G = 0$ (rigidity).

(iv) **First-order.** ρ_T involves one derivative of i; it detects how the structure *varies*, not just what it is at a point.

Remark 1 (The difficulty plane). The pair $(\rho_T, |\mu|)$—where $|\mu|$ is the Beltrami modulus—defines a two-dimensional "difficulty plane." The invariant $|\mu|$ measures algebraic deviation (anisotropy); ρ_T measures differential deviation (spatial variation). They are categorically independent.

2 The Burgers connection

The obstruction $G = i_x + i \cdot i_y$ has the algebraic form of the *Burgers nonlinearity* $u\, u_x$ in the inviscid Burgers equation $u_t + u\, u_x = 0$. This is not a coincidence: the VES monograph identifies G as the nonlinear term in the Burgers transport equation for the spectral parameter λ.

The Burgers equation is the gateway to a remarkable chain of connections.

2.1 The chain: Burgers → KPZ → Schur

Step 1. Burgers → KPZ. The Hopf–Cole transformation $u = -2\nu\,\partial_x \log Z$ linearises Burgers into the heat equation. Adding space-time white noise η gives the *KPZ equation* (Kardar–Parisi–Zhang, 1986):

$$\partial_t h = \nu\,\partial_x^2 h + \tfrac{\lambda}{2}(\partial_x h)^2 + \eta(x,t).$$

Setting $u = \lambda\,\partial_x h$ recovers stochastic Burgers; the Hopf–Cole transform $Z = \exp(\lambda h/2\nu)$ gives the stochastic heat equation with multiplicative noise, whose solution is a *directed polymer* partition function.

Step 2. KPZ → growth models. KPZ is the continuum limit of discrete growth models. The cleanest is *last passage percolation* (LPP): given i.i.d. weights $w(i,j)$ on $\mathbb{Z}^2_{>0}$, define

$$G(m,n) = \max_{\pi:(1,1)\to(m,n)} \sum_{(i,j)\in\pi} w(i,j)$$

over up-right lattice paths π. The surface $h(m,n) = G(m,n)$ grows in the KPZ universality class with fluctuation exponent $\beta = 1/3$.

The *totally asymmetric simple exclusion process* (TASEP)— particles on $\mathbb{Z}$ hopping right at rate 1 if the target is empty—is equivalent to LPP (Rost, 1981).

Step 3. Growth models → Schur functions (via RSK). The Robinson–Schensted–Knuth correspondence bijects nonneg-integer matrices to pairs of semistandard Young tableaux. Greene's theorem (1974): the shape λ_1 equals the longest weakly increasing subsequence.

For LPP with geometric weights $w(i,j) \sim \mathrm{Geom}(x_i y_j)$: the last passage time $G(m,n)$ equals λ_1 under RSK. The random partition is distributed according to the *Schur measure* (Okounkov, 2001):

$$\mathrm{Prob}(\lambda) \;\propto\; s_\lambda(x_1,\dots,x_m)\,s_\lambda(y_1,\dots,y_n),$$

derived from the Cauchy identity $\sum_\lambda s_\lambda(x)\,s_\lambda(y) = \prod_{i,j}(1-x_i y_j)^{-1}$.

Step 4. Schur measure → Tracy–Widom. The Schur measure is a *determinantal point process*. The probability $\mathrm{Prob}(\lambda_1 \le t)$ is a Fredholm determinant; steepest-descent analysis yields the *Airy kernel*, and

$$F_2(s) = \det(I - K_{\mathrm{Airy}})_{L^2(s,\infty)} = \exp\Big(-\int_s^\infty (x-s)\,q(x)^2\,dx\Big),$$

where q solves Painlevé II: $q'' = sq + 2q^3$. This is the *Tracy–Widom distribution*, first discovered as the fluctuation law for the largest eigenvalue of GUE random matrices (Tracy–Widom, 1994).

2.2 The role of Yang–Baxter and transfer matrices

The exact solvability at every level of this chain rests on the Yang–Baxter equation (YBE). For a vertex model with weights $R_{ab}^{cd}(u)$, the YBE ensures that transfer matrices $T(u) = \mathrm{tr}_{\mathrm{aux}} \prod_i R_i(u)$ commute: $[T(u), T(v)] = 0$. This is why:

- Schur functions are determinants (Jacobi–Trudi = Lindström–Gessel–Viennot for non-intersecting paths).
- The Schur measure is determinantal (commuting transfer matrices $\Rightarrow$ free fermions).
- The Fredholm determinant is computable in closed form.

The transfer operators T_{free} and $\bar{T}_{\mathrm{free}}$ of Zinn-Justin [11], acting on the fermionic Fock space of binary strings, sit precisely in this framework. The Littlewood–Richardson coefficients computed via puzzle pieces control the product structure of Schur functions, governing multi-point correlations in the Schur process.

3 Discretisation: from ρ_T to local rules

Observation 1. *Discretise Ω on a grid. At each vertex (x, y), place a generator $i(x, y) \in \mathbb{D}$. The rigidity condition $G = i_x + i \cdot i_y = 0$ becomes a* nearest-neighbour constraint*: the values of i at adjacent sites must satisfy a compatibility relation.*

This is exactly the setup for:

- **Puzzle pieces** (Knutson–Tao–Woodward): local rules constraining adjacent tiles.
- **Growth diagrams** (Fomin): local rules propagating partitions through a grid, where the local rule is a zero-curvature condition around each plaquette.
- **Vertex models** (six-vertex, ice model): local weights depending on adjacent edge states.

The generator $i \in \mathbb{D}$ plays the role of the local state; $G = 0$ plays the role of the local rule. The fermionic Fock space of binary strings (encoding partitions) is a discretisation of the space of generators.

Question 1. *Is there a natural discretisation of $G = 0$ on a triangular or square lattice that recovers the interlacing condition on partitions? If so, ρ_T would literally measure* how much interlacing fails*.*

3.1 The difficulty plane as (inhomogeneity, anisotropy)

In lattice model language, the two axes of the difficulty plane map to fundamental distinctions:

VES invariant	Lattice model	Combinatorial effect
$\lvert\mu\rvert$ (Beltrami)	Anisotropy of local weights	Deviation from isotropy
ρ_T (transport)	Inhomogeneity (weights vary site-to-site)	Richer combinatorics

Homogeneous isotropic models give product-formula partition functions. Homogeneous anisotropic models give the six-vertex model, Schur functions, determinantal formulas. Inhomogeneous models are harder but produce the richest combinatorics: factorial Schur functions, double Schubert polynomials.

4 Profile as discrete monodromy

This is the most speculative—and perhaps most suggestive— connection.

4.1 The analogy

In VES, ρ_T measures the transport obstruction: the monodromy of the complex structure under parallel transport. The condition $G = 0$ means trivial monodromy.

A cylindric partition on a cylinder of circumference d with profile $c = (c_0, c_1, \ldots, c_{r-1})$ has a *discrete monodromy*: as one traverses the cylinder, the boundary conditions shift by the profile.

VES (continuous)	**Cylindric partitions (discrete)**
$G = 0$, trivial monodromy	Trivial profile $\to$ plane partitions
$G \neq 0$, $\rho_T > 0$	Nontrivial profile $\to$ cylindric partitions
ρ_T measures twist magnitude	Profile measures boundary shift

Conjecture 1 (Informal)**.** *There exists a discretisation scheme under which the cylindric profile c arises as the holonomy of the discrete VES connection around the cylinder, and ρ_T (integrated around the cycle) recovers a function of c.*

4.2 Positivity as "mild deformation"

Under this analogy, Warnaar's positivity conjecture—that $Q_{n,c}(q) \geq 0$ coefficient-wise for rank-3 cylindric partitions—asserts that *even when $\rho_T > 0$, the partition function retains positivity.* The deformation from exact solvability is mild enough that combinatorial structure survives without catastrophic cancellation.

4.3 The $d \equiv 0 \pmod 3$ resonance

For rank 3, when $d \equiv 0 \pmod 3$ the profile distributes evenly among the three parts: the slack vector is $(d/3, d/3, d/3)$, which has no cyclic descent. The cyclic-descent injection rule (which works for all other d) fails here.

In the continuous picture, this corresponds to *resonant monodromy*: the monodromy divides evenly into the rank, creating a fixed point under the cyclic symmetry. In KAM theory, resonant frequencies are exactly where small-denominator problems arise and invariant tori break down. The $d \equiv 0 \pmod 3$ obstruction may be a discrete manifestation of the same phenomenon.

5 Poincaré factors and q-series

A more speculative thread. The Poincaré conformal factor at $z \in \mathbb{D}$ is $(1-|z|^2)^{-1}$. Imagine a sequence of generators at different scales with $|i_k| = q^k$. The product of Poincaré factors gives

$$\prod_{k=1}^{n} \frac{1}{1-q^{2k}},$$

which is a partition generating function (partitions into even parts $\leq 2n$). The full q-Pochhammer symbol $(q;q)_\infty = \prod_{n\geq 1}(1-q^n)$—ubiquitous in partition theory—arises from a product over all levels.

The Euler–Heine factorisation $\mathrm{GK}_c = H(z) \cdot F_c$, where $H = 1/(zq;q)_\infty$, decomposes the generating function into a "universal" part (product of Poincaré-like factors) and a "structural" part F_c. In the VES picture, this might correspond to:

universal hyperbolic metric contribution $\times$ structure-specific transport geometry.

Question 2. *Is there a multi-level VES (a tower of generators $i_1, i_2, \ldots$ with prescribed nesting) whose "total Poincaré factor" is the q-Pochhammer symbol? If so, the E–H factorisation would have a geometric interpretation as separation of metric and transport contributions.*

6 Open directions

1. **Construct the discretisation.** Make Section 3 precise: discretise the VES rigidity condition on a square lattice and identify the resulting local rule. Does it give interlacing? A vertex model?

2. **Deformed Yang–Baxter.** If ρ_T measures distance from exact solvability, what do Schur measures look like when Yang–Baxter is only approximately satisfied? Is there a "ρ_T-deformed" determinantal structure?

3. **Profile from holonomy.** Make the Conjecture in Section 4 precise: construct a discrete VES on a cylindrical lattice whose holonomy recovers the cylindric profile c.

4. **Resonance and positivity.** Can KAM-type arguments (small denominators at resonant frequencies) explain why $d \equiv 0 \pmod 3$ is harder for the positivity conjecture?

5. **Growth diagrams as flat connections.** Fomin's local rule for growth diagrams is a zero-curvature condition around each plaquette. Formalise the correspondence: local rule = discrete $G = 0$, and the failure $G \neq 0$ = "curvature" of the growth diagram.

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